A fixed-support exit from an internal pair equality

Human Compression project: L044 continuation

Retain the complete hypotheses of Theorem 4 of *Ambient transport and internal pair deficits*, including $k \notin F$, intransitivity of the full local $K = G_{>k}$ on the actual $H = G_{\geq k}$-orbit O, and equality $|F| = q - 1$ with $|O| = 2q$, $q \geq 3$. Thus $K|_O$ is the diagonal S_q on two natural rows, and every tail mark outside $F \cup \{k\}$ fixes O. Let $a = \min F$ and $x = \max F = a + q - 2$; here a denotes a label, not the crossing vertex used elsewhere. Put $P = G_{<k}$. Suppose also that a specified set $C \subseteq \mathrm{Fix}(K)$ has size $h \geq 1$. The ambient group G is transitive.

Lemma 1 (A third fixed row)**.** *Every point of the union $O^P = \bigcup_{p \in P} O^p$ is fixed by $G_{>x}$. This union is a disjoint union of K-orbits of size q, is disjoint from C, and has at least $3q$ points. Consequently*

$$|\,\mathrm{Fix}(G_{>x})| \geq 3q + h. \tag{1}$$

Proof. The two rows of O are actual K-orbits of size q. The prefix P centralizes K, so each of their translates is again a K-orbit of size q. Distinct orbits are disjoint and none contains a point of C, all of whose points are fixed by K. If $O^P = O$, then P preserves O; so does H, by its definition. Since $G = \langle P, H \rangle$, transitivity would give $O = \Omega$. This is impossible because C is nonempty and disjoint from O. Thus at least one further q-point K-orbit occurs in O^P.

Every mark after x lies outside $F \cup \{k\}$ and hence fixes O pointwise. Also $G_{>x} \leq K$, so it commutes with P. For $v \in O$, $p \in P$ and $g \in G_{>x}$, one has $(v^p)^g = (v^g)^p = v^p$. Thus $G_{>x}$ fixes O^P pointwise. It fixes C as a subgroup of K. The disjointness already proved gives (1). This argument does not require K to be transitive on any larger set or identify two distinct row orbits. □

Corollary 2 (The exact source exit)**.** *Put $X = \Omega \setminus \mathrm{Fix}(G_{>x})$ and $n = |\Omega|$. If*

$$2a \leq q + h + 3, \tag{2}$$

then

$$x \leq \frac{n - |X| - 1}{2}. \tag{3}$$

In particular this holds when the internal path starts at $k+1$ in either source head configuration $(k, h) = (2, 2)$ or $(5, 7)$. This is alternative (a) of source Proposition 5.31.

Proof. Since $x = a + q - 2$, condition (2) says $2x + 1 = 2a + 2q - 3 \leq 3q + h$. Lemma 1 gives $2x + 1 \leq n - |X|$, which is (3). If $a = k + 1$, condition (2) reduces to $q \geq 1$ for $(k, h) = (2, 2)$ and to $q \geq 2$ for $(k, h) = (5, 7)$; both follow from $q \geq 3$. There is no assumption $x < r - 1$: if $x = r - 1$, the fixed set in (1) is all of Ω and the same inequality remains valid. □

Source and scope. The target is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.31(a) and its internal-orbit cases, physical pages 38 and 42–43. Its two head configurations supply two or seven specified K-fixed points; an application must identify these actual points. The fixed-support target is borrowed; the third-row proof avoids discarding the even flip pattern. The full internal-pair proof and its ambient-eligibility hypotheses remain paid. A later path is covered only under (2). Other deficits, tables and full L044 remain pending; no computation, formal theorem or question credit is added.